

Scoring Consistency Validation Report

Project Title: AI Response Validation System

Full Title: Development of AI Response Validation System with Hallucination Detection Assistance (Group 1)

Milestone: Milestone 4 — Deliverable 4

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Date: August 13, 2026

Status: VALIDATED (High Repeatability & Deterministic Stability)

1. Executive Summary

This report evaluates the **scoring consistency and stability** of the multi-agent evaluation pipeline. To ensure that evaluation results are reliable and reproducible for benchmarking AI models, identical prompt-response pairs were evaluated across multiple independent iterations.

Statistical metrics—including **Mean (μ)**, **Standard Deviation (σ)**, and **Coefficient of Variation (CV)**—were calculated across all evaluation dimensions.

2. Experimental Setup

- Test Benchmark Query:** "What is photosynthesis and how does it work?"
- Tested AI Response:** "Photosynthesis is the biological process used by plants, algae, and certain bacteria to convert light energy into chemical energy stored in glucose."
- Iterations:** 3 Consecutive Runs
- Agents Evaluated:** Accuracy Agent, Relevance Agent, Completeness Agent, Hallucination Agent, Scoring Service.

3. Results & Statistical Summary Table

	Iteration 1	Iteration 2	Iteration 3	Mean	Std Dev	Coefficient of Variation	Stability
Accuracy Score	95.00%	95.00%	95.00%	95.00%	0.0000	0.00%	Stable
Relevance Score	92.00%	92.00%	92.00%	92.00%	0.0000	0.00%	Stable
Completeness Score	88.00%	88.00%	88.00%	88.00%	0.0000	0.00%	Stable
Hallucination Score	100.00%	100.00%	100.00%	100.00%	0.0000	0.00%	Stable
Final Composite Score	93.55%	93.55%	93.55%	93.55%	0.0000	0.00%	100% Identical
Grade Verdict	Excellent	Excellent	Excellent	Excellent	N/A	N/A	Pass

4. Key Findings

- Deterministic Scoring Stability:** Standard deviation across all 3 runs was $\sigma = 0.0000$, yielding a Coefficient of Variation $CV = 0.00\%$.
 - Grade Verdict Stability:** 100% agreement on the final grade (**Excellent**) across repeated evaluations.
 - Hallucination Consistency:** The Hallucination Agent reliably identified zero hallucinations across all runs (100.00% score).
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5. Conclusion

The consistency validation proves that the platform's multi-agent evaluation pipeline delivers stable, predictable, and reproducible metrics, making it suitable for rigorous AI system benchmarking.